What Do Source Models Buy Us When Predicting Full-Benchmark LLM Scores from a Few Items? A Controlled Study under Distribution Shift

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Abstract

Evaluating a language model on a full benchmark is costly, so many methods run a model on a small item subset and predict its full-benchmark score using item-level results of previously evaluated "source" models. We study this in a controlled factorial design on 13 binary item-level score matrices (Open LLM Leaderboard v2 and HELM): six item selectors (random, difficulty-stratified, anchor clusters, IRT-anchors, D-optimal, max-variance) crossed with plain-mean, regression, kNN and IRT-based estimators, under an IID split of models and two *extrapolation* splits where the models to be evaluated are more accurate than all sources. With 30 replicate splits per cell and bootstrap intervals, we find: under IID splits, model-based estimators reduce RMSE by 17–29% relative to a random-subset mean (k=50) and improve Spearman correlation by about 0.04, but selecting items by clustering or by information without a model-based estimator is at best marginally better than random (Anchor-IRT, 0.90) and otherwise worse; under extrapolation every method that relies on the source models for its point estimate is 15–140% *worse* than a random-subset mean, with a bias that grows with the target–source accuracy gap. A per-target two-parameter recalibration removes most of this bias but only reaches parity with the random mean (0.96), and with held-out model families but no strength shift model-based estimators work again. Methods that keep item sampling random are robust: source-difficulty stratification lowers RMSE by 12–16% in all regimes. We also study a cross-fitted finite-population difference estimator (CF), which we prove unbiased under simple random sampling and verify empirically (no detectable bias; 92–94% coverage of Wald intervals), and an empirical-Bayes cohort shrinkage variant (CF-EB) that gets 25% lower RMSE than random in IID and 12–16% under extrapolation, at the price of a small bias. No estimator improved the rank correlation under extrapolation by more than 0.005 at $k \geq 50$ (at most 0.04 at smaller k). Gains of safe methods are modest, and we discuss the limits of this evidence.

1. Introduction

Evaluating a language model on a modern benchmark means answering thousands of items; repeating it for many checkpoints, prompts or inference settings is expensive. A tempting shortcut is to run a model on a small subset and extrapolate to the full benchmark, using item-level results of models that have already been evaluated ("source models") to choose the subset and to calibrate the extrapolation. Several recent methods do this with clustering, item response theory (IRT) or learned selection policies (Vivek et al., 2024; Maia Polo et al., 2024; Kipnis et al., 2025; Li et al., 2025; Pacchiardi et al., 2024).

What should we expect from such methods in the situation where they are most needed, i.e. when the model under test is *new* and different from the source models? Two issues make this setting hard. First, a method is typically validated on a random split of models, where target models resemble sources. Second, any dependence on the source models enters through two channels—*which items are kept* and *how the subset is turned into a score*—and these are rarely separated, so it is unclear which channel brings the gains or the failures.

We study this with a controlled factorial design on 13 binary item-level score matrices (Open LLM Leaderboard v2 and HELM, up to 429 models and 12,032 items). We cross six item selectors with several estimators, evaluate under four protocols (IID model split, two *extrapolation* splits in which targets are the more accurate models, and a held-out-family split), report MAE, RMSE, Spearman and Kendall with bootstrap intervals over benchmarks and replicates, and measure bias and interval coverage with re-

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peated subsampling. Because plain model-based estimators turn out to be strongly biased under extrapolation, we additionally study a per-target recalibration of the predictor, a cross-fitted, finite-population difference estimator (CF) that is exactly unbiased under random item sampling, and an empirical-Bayes variant (CF-EB) that shrinks towards the plug-in when the cohort's own data show the plug-in is accurate.

Contributions. (1) A factorial, paired comparison of item selectors and estimators with uncertainty quantification, separating the effect of selection from that of estimation and the IID from the extrapolation regime (Section 4). (2) A cross-fitted difference estimator with a proof of exact unbiasedness under random item sampling, design-based confidence intervals, and a cohort-adaptive shrinkage version (Section 3). (3) Evidence on when each family works and a list of failure modes, including negative results (Sections 4 and 5). All code and the pinned data version are provided for reproduction.

2. Related Work

Selecting items. Anchor Points (Vivek et al., 2024) clusters items by the responses of many models and evaluates one representative per cluster; tinyBenchmarks (Maia Polo et al., 2024) clusters IRT item embeddings and adds IRT-based estimators (p-IRT, gp-IRT); metabench (Kipnis et al., 2025) keeps the most IRT-informative items and predicts scores with additive models; Li et al. (2025) learn a policy that selects items adaptively and imputes the rest; Pacchiardi et al. (2024) train an "assessor" on a handful of reference instances of the new model and report that random reference instances work about as well as the more elaborate selectors they tried, which agrees with part of our findings. **Prediction under shift.** Zhang et al. (2025) compare eleven benchmark-prediction methods on 19 benchmarks, observe that a random sample followed by regression is competitive, that methods degrade when the new model is more accurate than the source models, and propose an estimator inspired by augmented inverse-propensity weighting. Our study is complementary to that work (of which we read only the abstract, and we did not use its code): we use the (public) score matrices they release, but design our own factorial comparison (6 selectors $\times$ several estimators), define the shift regimes through the true accuracy ranking of the models, quantify unbiasedness and interval coverage by repeated subsampling, and study a cross-fitted finite-population difference estimator with a cohort-adaptive shrinkage step. We do not claim priority for the idea of correcting a model-based estimate by sample residuals: it is classical in survey sampling (model-assisted/GREG estimation, Särndal et al. 1992) and underlies prediction-powered inference (Angelopoulos et al., 2023) and double machine learning (Chernozhukov et al., 2018); the novelty is the exact finite-population form with out-of-fold residuals for item sampling, the empirical-Bayes shrinkage across a cohort of models sharing the sample, and the systematic empirical comparison. **Evaluation data.** Our data are item-level correctness matrices derived from the Open LLM Leaderboard (Fourrier et al., 2024) (including MMLU-Pro, Wang et al. 2024) and HELM (Liang et al., 2023), as packaged by Zhang et al. (2025).

3. Problem Setting, Methods and Protocol

Setting. A benchmark has n items. We have a binary *source matrix* $X \in \{0,1\}^{m\times n}$ with the item-level correctness of m source models. A *cohort* of T target models is evaluated on a subset $S \subset [n]$ of $k \ll n$ items, giving $Y_{t,S}$, and the goal is to predict each target's full-benchmark accuracy $\theta_t = \frac{1}{n}\sum_{i=1}^{n} y_{ti}$. Metrics are computed over the cohort against the true θ_t: MAE, RMSE, signed bias, Spearman ρ and Kendall τ. A *method* is a pair (selector, estimator); the selector sees only X (and may use randomness), the estimator sees X, S and $Y_{\cdot,S}$. Evaluating all targets on the same S mirrors leaderboard or checkpoint-comparison use.

3.1. Item selectors

Random (simple random sampling, SRS). **Stratified** (a standard survey-sampling device that we add as a baseline): items are sorted by their source-model accuracy and sampled systematically with a random start; every item has inclusion probability k/n, so the sample is spread evenly over the source-difficulty range. **Anchor points** (Vivek et al., 2024): k-means over items, each item represented by its response vector over source models (reduced to a 50-dimensional SVD embedding for speed); the item closest to each centroid is the anchor, weighted by cluster size. **Anchor-IRT** (Maia Polo et al., 2024): the same, but clustering the IRT item parameters. **D-optimal**: greedy selection maximising $\log\det$ of the Fisher information for the ability vector under the fitted IRT model, averaged over source abilities (an information-based design in the spirit of Kipnis et al. (2025)). **Max-variance**: the k items whose source accuracy is closest to 0.5.

3.2. Latent-factor (IRT) predictor

All model-based estimators share one predictor: a multidimensional two-parameter logistic IRT model $P(y_{ti} = 1) = \sigma(\beta_i + \alpha_i^\top \vartheta_t)$ with $\vartheta_t \in \mathbb{R}^d$ (Lord & Novick, 1968; Reckase, 2009; Maia Polo et al., 2024). Item parameters (α_i, β_i) and source abilities are fit by MAP (L2 priors, L-BFGS) on X only. The latent dimension $d \in \{1,2,4,8\}$ is chosen per split by the held-out log-loss of a quarter of the *source* models (never using target data). For a target, $\hat{\vartheta}_t$ is the MAP estimate given $Y_{t,S}$ (standard-normal prior), and

$\hat{p}_{ti} = \sigma(\beta_i + \alpha_i^\top \hat{\vartheta}_t)$.

3.3. Estimators

Mean. $\hat{\theta}_t = \sum_{i \in S} w_i y_{ti}$ ($w_i = 1/k$, or cluster-size weights for anchors). **Direct regression.** Ridge regression from the k subset outcomes to the full score trained on the source models (the "random sample + regression" baseline family, Zhang et al. 2025), and a kNN estimator averaging the full scores of the 10 source models closest on S. **p-IRT / gp-IRT** (Maia Polo et al., 2024): $\hat{\theta}_t^{\text{p-IRT}} = \frac{k}{n} \sum_{i \in S} w_i y_{ti} + \frac{n-k}{n} \overline{\hat{p}_{t,\bar{S}}}$ (observed items plus mean prediction on the rest); gp-IRT mixes this with the weighted mean using the weight $\lambda = \hat{b}^2/(\hat{\sigma}^2/k + \hat{b}^2)$, where $\hat{b}$ is an IRT-bias estimate computed on held-out *source* models and $\hat{\sigma}^2$ the mean Bernoulli variance (divided by 4 for anchor selectors, as in the original recipe).

3.4. Cross-fitted model-assisted estimators (CF, CF-EB)

Plug-in estimators inherit the bias of $\hat{p}$ when the target is unlike the source models. Survey sampling offers a remedy: with a *randomised* S, a difference estimator corrects the model-based prediction by the average residual on the sample (Särndal et al., 1992), which is unbiased whatever the quality of the model (cf. prediction-powered inference, Angelopoulos et al. 2023). Because $\hat{p}_{t\cdot}$ is fit on the same sampled items, the residuals must be computed out of fold (cross-fitting, Chernozhukov et al. 2018). Split S uniformly at random into K folds $S_1, \ldots, S_K$ (we use $K = 5$); let $T_j = S \setminus S_j$ and let $\hat{p}_t^{(-j)}$ be the predictions from the IRT ability estimated on T_j only. The fold-j estimator is

$$\hat{\theta}_t^{(j)} = \frac{1}{n}\Big[\sum_{i \in T_j} y_{ti} + \sum_{i \notin T_j} \hat{p}_{ti}^{(-j)}\Big] + \frac{n - |T_j|}{n} \cdot \frac{1}{|S_j|} \sum_{i \in S_j} \big(y_{ti} - \hat{p}_{ti}^{(-j)}\big), \tag{1}$$

and $\hat{\theta}_t^{\text{CF}} = \frac{1}{K}\sum_j \hat{\theta}_t^{(j)}$. We write $\hat{\theta}_t^{\text{plug}}$ for the average of the first (plug-in) term over folds, i.e. the cross-fitted plug-in, and $\hat{\delta}_t$ for the averaged residual term, so $\hat{\theta}_t^{\text{CF}} = \hat{\theta}_t^{\text{plug}} + \hat{\delta}_t$.

Proposition 1 (Unbiasedness)**.** *If S is a simple random sample of size k from the n items and the folds are a uniformly random partition of S, then $\mathbb{E}[\hat{\theta}_t^{\text{CF}}] = \theta_t$ for every target t, every item-parameter estimate (α, β) computed from X alone, and every (possibly wrong) latent model. The expectation is over the draw of S and the fold assignment, with the target's responses fixed.*

Proof sketch. Given T_j (its items and outcomes), S_j is a uniform random subset of the $n - |T_j|$ remaining items and $\hat{p}^{(-j)}$ is a fixed function of T_j (and X), so $\mathbb{E}[\text{mean residual on } S_j \mid T_j]$ equals the mean residual over the remaining items; adding the observed part $\sum_{T_j} y$ and the predictions on the remaining items recovers $\sum_i y_{ti}$ exactly. Full proof in Appendix A. □

Two further facts are used. (i) A Wald interval $\hat{\theta}_t^{\text{CF}} \pm 1.96\, \hat{s}_t \sqrt{(1 - k/n)/k}$, with $\hat{s}_t$ the standard deviation of the cross-fitted residuals, is a design-based confidence interval whose empirical coverage we test. (ii) CF is only as good as its variance: it removes the item-sampling variance that $\hat{p}$ explains but retains the Bernoulli noise of the sampled items, so gains are bounded by $1 - R^2$ of the residual variance (Section 4).

Cohort-adaptive shrinkage (CF-EB). Unbiasedness has a price: when the plug-in is nearly unbiased (targets similar to sources), the residual term only adds noise. Because a cohort of targets shares the same S, we can estimate how biased the plug-in is *for this cohort from the cohort's own subset data*. Let v_t be the estimated sampling variance of $\hat{\delta}_t$. Treating the plug-in's bias b_t as $b_t \sim \mathcal{N}(0, \tau^2)$ and $\hat{\delta}_t \mid b_t \sim \mathcal{N}(b_t, v_t)$, empirical Bayes (Efron & Morris, 1973) gives

$$\hat{\theta}_t^{\text{CF-EB0}} = \hat{\theta}_t^{\text{plug}} + \frac{\hat{\tau}^2}{\hat{\tau}^2 + v_t}\hat{\delta}_t, \qquad \hat{\tau}^2 = \max\Big(0, \tfrac{1}{T}\textstyle\sum_{t'} \hat{\delta}_{t'}^2 - \tfrac{1}{T}\sum_{t'} v_{t'}\Big). \tag{2}$$

If the plug-in is systematically biased (extrapolation), $\hat{\tau}^2 \gg v_t$ and the estimator tends to CF; if the plug-in is accurate, $\hat{\tau}^2 \approx 0$ and it tends to the plug-in. This is not unbiased (it trades bias for variance, as any shrinkage) and requires a cohort of $T \gtrsim 10$ models sharing S. A variant CF-EB1 shrinks towards a cohort-level linear trend of $\hat{\delta}_t$ on $\hat{\theta}_t^{\text{plug}}$ instead of towards zero. When the selector is stratified, Equation (1) treats the sample as equal-probability; this is exact for SRS and only approximate for systematic sampling, which we check empirically.

Recalibrated predictor. The IRT predictor above fixes how a target's success depends on source-estimated difficulty β_i. A natural remedy for a target with a different ability range is a per-target recalibration, i.e. a logistic fit with a free intercept a_t and difficulty slope δ_t: $\sigma(\beta_i + a_t + \delta_t \beta_i + \alpha_i^\top \vartheta_t)$ with ridge penalties 0.1 on a_t, 1 on δ_t and ϑ_t (a pilot over four penalty settings on three benchmarks, in the Extrap-50 regime, chose these before the main runs; recalibrated results may therefore be mildly optimistic). We use it in p-IRT ("recalibrated p-IRT"), in CF and in CF-EB (suffix "recal."). It costs two extra parameters per target.

3.5. Evaluation protocol

For each benchmark we draw 30 independent replicate splits of the models into source and target sets, under four regimes. **IID**: random 50%/50% split of models. **Extrap-50**: models are ranked by true full-benchmark accuracy; 80% (random)

of the bottom half are sources and *all* of the top half are targets. **Extrap-25**: sources are 80% of the bottom 75%, targets the top 25%. These mimic the practical situation in which the model to be evaluated is newer and better than any model with historical item-level results. **Family hold-out** (Open LLM Leaderboard benchmarks only, 7 of 13): organisations (the prefix of the model name) are drawn at random as targets until they make up $\geq 30\%$ of the models; all other models are sources. This tests novelty of the model family without a systematic strength shift. In each replicate we fit all source-side components, then for each $k \in \{10, 25, 50, 100, 200\}$ ($k \leq n/2$) apply each selector once and all estimators to the same S, so that comparisons are paired. We report accuracies in percentage points. As the headline we use the *relative RMSE*: the geometric mean over benchmarks of RMSE(method)/RMSE(Random+mean), each RMSE averaged over replicates; < 1 is better than the random-subset average (a ratio r is equivalent to using $1/r^2$ times as many random items). Confidence intervals are 95% percentile intervals of a two-level bootstrap (benchmarks, then replicates within benchmark); significance across benchmarks uses the Wilcoxon signed-rank test.

4. Experiments

Data and setup. We use the item-level 0/1 correctness matrices released with Zhang et al. (2025) (pinned commit, MIT licence): seven Open LLM Leaderboard v2 benchmarks (ARC-C, BBH, GPQA, IFEval, MMLU-Pro, MuSR, MATH Level 5; 212–429 models, 541–12,032 items) and six HELM benchmarks (79–83 models, 437–2,047 items); duplicated model rows (identical outcomes) are removed (Table 3, Appendix B). Each (benchmark, regime) cell uses 30 replicate splits, $k \in \{10, 25, 50, 100, 200\}$. Everything on the source side (IRT fit, d, gp-IRT constants, ridge, kNN) uses source models only. Unless stated otherwise we report $k = 50$ (all k in Appendix D).

4.1. Main results

Table 1 and Figure 1 summarise the comparison; per-benchmark values are in Figure 4 and Appendix D.

IID split: source models help; informed selection mostly does not. Every model-based estimator reduces error relative to a random-subset mean at $k = 50$ (ridge 0.71, gp-IRT 0.78, p-IRT 0.83; Wilcoxon over benchmarks $p < 0.01$ except p-IRT, $p = 0.07$) and raises Spearman from 0.858 to 0.89–0.90. The benefit shrinks with k (Figure 1): ridge reaches 0.56 at $k = 10$, and at $k = 200$ the model-based and randomised methods are all within 0.80–0.92. Choosing items by clustering or by information *without* a model-based estimator does not help: anchors with a weighted mean are worse than random (1.13), Anchor-IRT is slightly better (0.90), and D-optimal / max-variance subsets are strongly biased (2.5–3.0× the error), because items near 50% source accuracy or with large loadings are unrepresentative of the whole benchmark. Selection does matter when paired with the matching estimator: the tinyBenchmarks-style pairing (Anchor-IRT + gp-IRT, 0.71) beats Random + gp-IRT (0.78), whereas D-optimal items + p-IRT (1.00) are no better than random. The only selector that helps with a plain mean in every regime is the *difficulty-stratified* sample (0.84; better on 13/13 benchmarks), which stays a random sample.

Extrapolation: model-based estimators and informed subsets fail; randomisation protects. When targets are more accurate than all sources, *every* method that relies on the source models for the point estimate is worse than the plain random-subset mean at $k = 50$: ridge 1.82, kNN 2.43, p-IRT 1.66, gp-IRT 1.42, Anchor-IRT + gp-IRT 1.22 (Extrap-50; 1.41, 1.92, 1.61, 1.42, 1.15 in Extrap-25), and Spearman falls from 0.73 to 0.38–0.70. The signed-error plot (Figure 2) shows why: p-IRT is nearly unbiased for weak targets but under-predicts strong targets by up to 13 points, and the bias grows with the accuracy gap between targets and sources (Figure 3; Pearson $r = -0.62$ over 39 benchmark-regime cells). The bias is not explained by items that no source model solves (the share of target successes on such items is below 1% in 36 of 39 cells; exceptions: IFEval and MATH-L5). Deterministic selectors (anchors, D-optimal, max-variance) with a plain mean are biased by design (up to +16 points for max-variance, Appendix D). By contrast, methods that keep item sampling random hold up: the stratified mean (0.88/0.85 in Extrap-50/25), CF (0.90/0.86) and CF-EB0 (0.88/0.84) beat the random mean in 32–38 of the 39 (benchmark, $k \in \{25, 50, 100\}$) cells (e.g. 36/39 for CF in Extrap-50; per-k Wilcoxon $p \leq 0.01$). At very small k the shrinkage estimators are best (CF-EB0 0.80/0.78 at $k = 10$ vs 0.93/0.89 for CF) since the sample mean is then very noisy.

A two-parameter recalibration removes most of the bias but not the variance problem. Letting each target have its own intercept and difficulty slope in the IRT predictor reduces p-IRT's extrapolation bias from -8.6 to -0.5 points and its relative RMSE from 1.66 to 0.96 [0.86, 1.12] (Extrap-50; 1.61 → 0.96 in Extrap-25), i.e. to parity with the random mean (better on 32–34 of 39 cells but $p = 0.15/0.03$) with a lower Spearman (0.701 vs 0.733; 0.568 vs 0.655 in Extrap-25). This is a strong baseline that our first version of the study lacked; in IID it gives 0.86 (vs 0.83 for plain p-IRT). It has no unbiasedness guarantee and no interval.

Family hold-out: model-based estimators work. When whole organisations are held out but accuracy is not shifted, source-informed estimators are competitive again (ridge 0.60, gp-IRT 0.68, p-IRT 0.69, CF-EB0 0.69, CF 0.82; recalibrated p-IRT 0.87), so the failures above are due to the

Table 1. Relative RMSE (geometric mean over benchmarks of RMSE divided by that of Random+mean; < 1 is better than a random subset; 95% bootstrap CI over benchmarks and replicates) and Spearman ρ between predicted and true scores of the target models, $k = 50$. IID: random model split. Extrap-50/25: targets are the top 50%/25% most accurate models. Family: whole organisations held out (7 Open LLM Leaderboard benchmarks only, hence a different average). "recalibrated": per-target intercept and difficulty slope (Section 3).

	IID		Extrap-50		Extrap-25		Family	
Method (selector + estimator)	rel. RMSE	ρ	rel. RMSE	ρ	rel. RMSE	ρ	rel. RMSE	ρ
Random + mean	1	0.858	1	0.733	1	0.655	1	0.805
Stratified (difficulty) + mean	0.84 [0.80,0.88]	0.864	0.88 [0.84,0.93]	0.737	0.85 [0.81,0.89]	0.658	0.86 [0.80,0.91]	0.802
Anchor points + weighted mean	1.13 [0.96,1.32]	0.859	1.69 [1.29,2.21]	0.706	1.45 [1.21,1.72]	0.599	1.18 [0.94,1.52]	0.787
Anchor-IRT + weighted mean	0.90 [0.85,0.95]	0.864	1.06 [0.92,1.30]	0.701	0.96 [0.86,1.10]	0.632	0.90 [0.85,0.95]	0.804
D-optimal + mean	2.51 [1.93,3.34]	0.893	2.70 [2.08,3.45]	0.588	2.79 [2.14,3.75]	0.480	2.83 [1.98,4.75]	0.858
Max-variance + mean	2.95 [2.17,4.24]	0.917	2.96 [2.28,4.02]	0.599	2.52 [1.77,3.66]	0.559	3.25 [2.09,5.73]	0.905
Random + ridge regression	0.71 [0.58,0.84]	0.904	1.82 [1.50,2.22]	0.518	1.41 [1.18,1.65]	0.514	0.60 [0.48,0.76]	0.881
Random + kNN over models	1.01 [0.73,1.37]	0.884	2.43 [1.86,3.15]	0.377	1.92 [1.51,2.41]	0.356	0.70 [0.52,0.99]	0.859
Random + p-IRT	0.83 [0.69,0.97]	0.892	1.66 [1.29,2.13]	0.545	1.61 [1.26,2.07]	0.513	0.69 [0.56,0.87]	0.852
Random + gp-IRT	0.78 [0.67,0.90]	0.895	1.42 [1.15,1.75]	0.666	1.42 [1.14,1.79]	0.590	0.68 [0.55,0.85]	0.856
Anchor-IRT + gp-IRT (tinyBench.-style)	0.71 [0.62,0.80]	0.904	1.22 [1.05,1.42]	0.699	1.15 [0.97,1.38]	0.622	0.65 [0.56,0.76]	0.861
Random + p-IRT (recalibrated)	0.86 [0.81,0.93]	0.862	0.96 [0.86,1.12]	0.701	0.96 [0.83,1.21]	0.568	0.87 [0.78,1.00]	0.815
Random + CF	0.83 [0.80,0.87]	0.867	0.90 [0.86,0.94]	0.734	0.86 [0.81,0.92]	0.653	0.82 [0.78,0.88]	0.817
Stratified + CF	0.82 [0.78,0.86]	0.871	0.88 [0.84,0.93]	0.735	0.84 [0.80,0.89]	0.653	0.83 [0.78,0.88]	0.814
Random + CF-EB0	0.75 [0.64,0.85]	0.892	0.88 [0.82,0.94]	0.727	0.84 [0.77,0.93]	0.652	0.69 [0.56,0.87]	0.853
Stratified + CF-EB0	0.73 [0.63,0.84]	0.897	0.86 [0.80,0.93]	0.728	0.83 [0.76,0.90]	0.650	0.69 [0.56,0.85]	0.850
Random + CF-EB1	0.74 [0.66,0.81]	0.892	0.86 [0.77,0.95]	0.655	0.79 [0.69,0.90]	0.574	0.69 [0.60,0.80]	0.852

strength shift, not to model novelty as such (7 benchmarks only).

Ranking. In extrapolation none of the methods improves Spearman over the random mean (0.733 in Extrap-50) by more than 0.005 at $k \geq 50$ (at most 0.02 at $k = 25$; anchors $+0.04$ at $k = 10$): CF 0.734, CF-EB0 0.727, stratified 0.737; CF-EB1 *hurts* ranking (0.655) although it lowers RMSE, since its linear-trend shrinkage compresses differences between targets. The RMSE gains of the safe methods are thus better calibration of scores, not better rankings.

Cohort-adaptive shrinkage keeps the IID gains without losing robustness. CF alone is unbiased but gains only 10–17% in RMSE (0.83 IID, 0.90/0.86 extrapolation) because the Bernoulli noise of the sampled items remains. CF-EB0 gets 0.75 in IID (CF 0.83, cross-fit plug-in alone 0.88) and 0.88/0.84 under extrapolation, close to the best of the two behaviours in each regime. It is, however, biased (mean signed error ≈ -1.7 points under extrapolation, Appendix D) and requires a cohort of models evaluated on the same subset. Stratifying and then applying the CF family gives small additional gains (0.73 vs 0.75 in IID), since stratification and the IRT predictor exploit the same source-difficulty information. With the recalibrated predictor, CF-EB0 behaves like CF (0.83/0.87/0.84) because recalibrated plug-ins carry little bias to shrink.

4.2. Unbiasedness and interval coverage

We verified Proposition 1 empirically: for 13 benchmarks $\times$ 2 regimes $\times$ 5 splits we drew 200 independent random subsets per split and compared each estimator with the truth (Table 2). CF shows no bias (mean error ≤ 0.07 points; at most 1% of target models have a bias z-score above 3), including under extrapolation, whereas the cross-fit plug-in and p-IRT are significantly biased for 83–96% of targets. CF has 10–20% smaller RMSE than the random mean at the same k in this larger check (geometric-mean ratio 0.80–0.90), and its Wald intervals are 10–20% shorter with empirical coverage of 92–94% (nominal 95%) versus 89–94% for the plain mean with its own Wald interval: item-level uncertainty is reported honestly for CF, but intervals are slightly anti-conservative at $k = 25$.

Table 2. Bias, RMSE (points) and 95% Wald coverage / mean width (points) over 200 random subsets $\times$ 13 benchmarks $\times$ 5 splits. |bias|: mean absolute per-target bias; sig.: share of targets with $|z| > 3$.

reg.	k	estimator	bias	|bias|	sig.	RMSE	cov.%	width
iid	25	Random mean	+0.02	0.48	0%	8.45	89.4	32.1
iid	25	CF	+0.04	0.39	0%	7.09	92.6	27.3
iid	25	cross-fit plug-in	-1.06	4.96	91%	6.89	–	–
iid	25	p-IRT	-0.92	4.39	89%	6.48	–	–
iid	100	Random mean	-0.01	0.22	0%	4.03	93.2	15.5
iid	100	CF	+0.00	0.18	0%	3.23	93.9	12.4
iid	100	cross-fit plug-in	-0.41	2.27	86%	3.59	–	–
iid	100	p-IRT	-0.34	2.01	83%	3.39	–	–
up50	25	Random mean	+0.11	0.48	1%	8.39	90.4	32.3
up50	25	CF	+0.07	0.43	1%	7.46	92.2	28.9
up50	25	cross-fit plug-in	-10.50	10.86	96%	12.25	–	–
up50	25	p-IRT	-9.90	10.18	96%	11.66	–	–
up50	100	Random mean	+0.03	0.24	0%	4.02	94.0	15.7
up50	100	CF	-0.01	0.20	0%	3.43	94.4	13.4
up50	100	cross-fit plug-in	-7.15	7.23	96%	8.23	–	–
up50	100	p-IRT	-6.69	6.75	95%	7.78	–	–

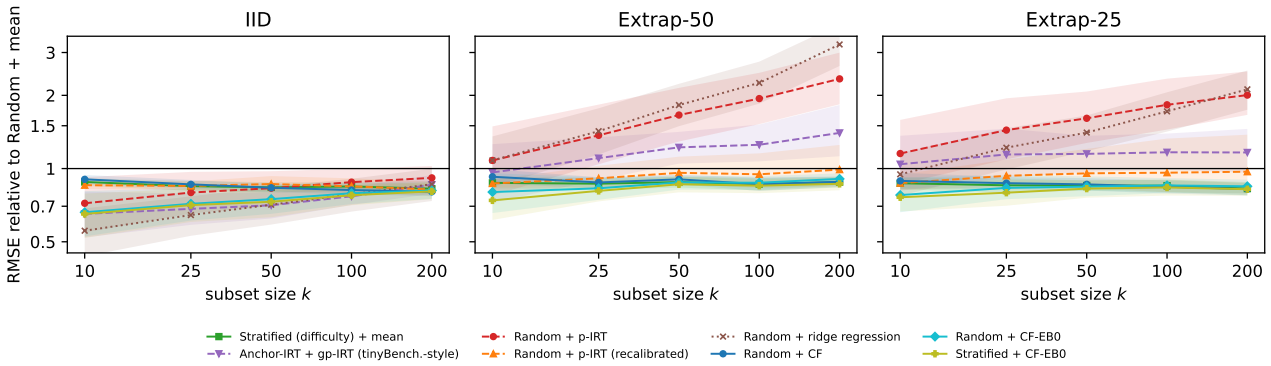


Figure 1. RMSE relative to Random+mean versus subset size k (geometric mean over 13 benchmarks; bands: 95% bootstrap CI). Plain model-based estimators gain under an IID split but lose increasingly under extrapolation; randomised source-informed methods stay at 0.8–0.9.

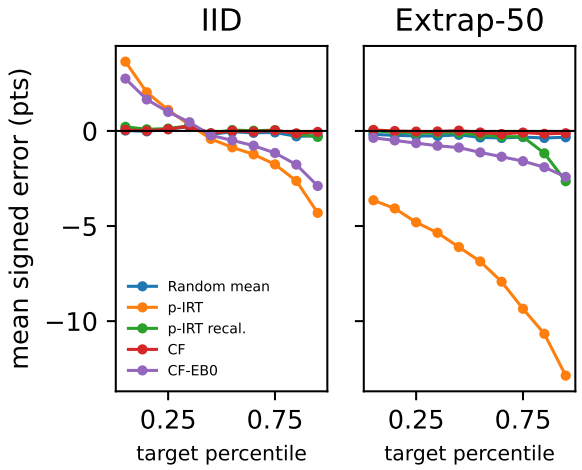


Figure 2. Mean signed error ($k = 100$, random subsets) by percentile of target strength (IID and Extrap-50), averaged over benchmarks. p-IRT regresses towards the source mean; CF is unbiased across strengths; CF-EB0 interpolates; recalibration removes most of the bias.

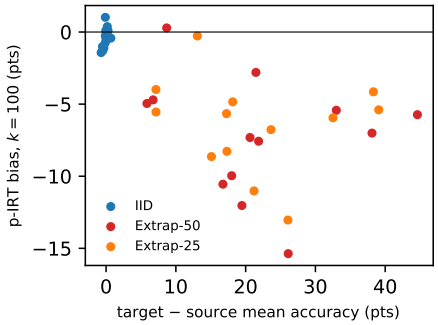


Figure 3. Bias of p-IRT ($k = 100$) versus the accuracy gap between target and source models; each point is one benchmark-regime cell (mean over 30 splits).

4.3. Ablations

Details are in Appendix E (10 replicates per cell; relative RMSE at $k = 50$ against Random+mean on the same draws). **Number of source models.** With few sources the IRT plug-in is unreliable even IID: p-IRT is worse than random with 8 or 16 source models (1.43, 1.22) and breaks even near 32 (1.03), reaching 0.72 with 128. CF stays at 0.83–0.89 and CF-EB0 at 0.67–0.84 for all m; under extrapolation p-IRT is 1.3–1.9 while CF stays at 0.91–0.95. **Latent dimension and ability prior.** A weaker prior on the target ability ($\lambda_\vartheta = 0.1$ instead of 1) improves p-IRT in both regimes (IID 0.80 $\rightarrow$ 0.69, Extrap-50 1.67 $\rightarrow$ 1.31 at $d = 4$), so the main-experiment p-IRT is not the best-tuned plug-in, but it does not cross 1 under extrapolation (1.21–1.62 across d); CF is insensitive. CF-EB0 with the weak prior is worse under extrapolation (0.96–0.98 at $d \geq 4$). **Folds.** $K \in \{2, 5, 10\}$ gives the same CF accuracy (0.82–0.84 IID, 0.90 Extrap-50). **Regression estimators.** Classical regression (GREG) estimators with source difficulty (or IRT item features) as covariates give 0.86/0.91 (0.86/0.93) in IID/Extrap-50 and add nothing to stratification or CF (0.82–0.83/0.89–0.91).

5. Discussion and Limitations

Take-aways. (i) The value of source-model information depends on whether the model under test resembles the sources *in strength*. Under an IID model split, IRT/regression estimators cut RMSE by 17–29% at $k = 50$ and give better rankings; under extrapolation the same estimators are 15–140% *worse* than a random-subset mean, with errors that grow with the accuracy gap; with held-out model families but no strength shift they work well again. A validation that splits models at random therefore says little about the intended use (evaluating a newer, better model). (ii) Where the source models are used only to *choose* items, informed

deterministic subsets do not reliably beat random sampling with a plain mean (Anchor-IRT: 0.90 IID but 1.06/0.96 under extrapolation; anchors, D-optimal and max-variance are worse than random); only source-difficulty *stratification*, which remains a random sample, consistently does (12–16% lower RMSE in every regime). (iii) A per-target recalibration (two parameters) removes most of the plug-in's extrapolation bias and brings p-IRT to parity with the random mean (0.96), but no better, and with worse rankings. (iv) A cross-fitted difference estimator makes model assistance safe (unbiased, calibrated intervals), but the variance reduction is modest (10–17%) since most residual variance is item-level Bernoulli noise; a cohort-adaptive shrinkage step recovers most of the IID gain (RMSE -25% at $k = 50$, -34% at $k = 10$) while staying at the level of CF under extrapolation, at the cost of a small bias. (v) No estimator improved *rankings* under extrapolation by more than 0.005 Spearman at $k \geq 50$.

Relation to prior findings. Our observations that random sampling is competitive as a selector and that prediction degrades for models more accurate than the sources agree qualitatively with Pacchiardi et al. (2024) and Zhang et al. (2025); our contribution is the controlled separation of selection and estimation, the bias diagnostics, the exact-unbiasedness/coverage verification, and the cohort-adaptive shrinkage. We did not run the authors' released code, so we do not claim to reproduce their numbers; our IRT-based baselines are our own implementations following the descriptions in Maia Polo et al. (2024) and differ from the released tinyBenchmarks code (e.g. our anchor clustering uses an SVD embedding, our IRT fit is a MAP fit with fixed priors).

Limitations. *Protocol.* Extrapolation is simulated by ranking models by their own full-benchmark accuracy; real shifts (new architectures, training data, families) may differ and no release dates were available. Model families are controlled only in the family hold-out regime (7 benchmarks). The target cohort is evaluated on one shared subset; CF-EB needs such a cohort ($T \gtrsim 10$) and is not applicable to a single model. *Methods.* Baselines are our re-implementations and use a single IRT family; stronger learned predictors or different selectors (e.g. adaptive acquisition, Li et al. 2025) were not tested, and we did not tune baselines per benchmark. CF is exactly unbiased only for simple random sampling; for the stratified sample the equal-probability formula is approximate (its mean signed error at k=50 is at most 0.13 points in all regimes, but we did not study its variance estimator or coverage). The recalibration penalties were set in a small pilot on three benchmarks in the Extrap-50 regime, so recalibrated numbers may be mildly optimistic; baselines were otherwise not tuned per benchmark (a weaker ability prior improves p-IRT, Section 4.3). *Evidence.* Benchmarks are few (13) and heterogeneous; the confidence intervals mix between-benchmark and between-split variation, and replicates share the same pool of models, so they do not capture uncertainty about the population of future models. Data are binary correctness from a fixed set of public leaderboards (possibly contaminated or containing models with degenerate outputs); we did not study continuous metrics, multi-benchmark pooling, or cost-based stopping rules. *Generality.* The relative gains we report (about 10–15% for safe methods) are modest: unbiased subset estimation cannot go much below the Bernoulli noise of k items, so in extrapolation the practical advice is to use more items or to treat model-based predictions as unreliable unless corrected.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning, specifically the cost-efficient and reliable evaluation of language models. Cheaper evaluation lowers the compute and energy cost of model development, but unreliable shortcut evaluation could also give a false sense of a model's capability; our results quantify when such shortcuts fail (notably for models stronger than those seen before), and we recommend unbiased, uncertainty-aware estimators in that regime. We see no further societal consequences that must be specifically highlighted.

References

Angelopoulos, A. N., Bates, S., Fannjiang, C., Jordan, M. I., and Zrnic, T. Prediction-powered inference. *arXiv preprint arXiv:2301.09633*, 2023.

Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., and Robins, J. Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68, 2018.

Efron, B. and Morris, C. Stein's estimation rule and its competitors—an empirical Bayes approach. *Journal of the American Statistical Association*, 68(341):117–130, 1973.

Fourrier, C., Habib, N., Lozovskaya, A., Szafer, K., and Wolf, T. Open LLM leaderboard v2. Hugging Face, 2024.

Kipnis, A., Voudouris, K., Schulze Buschoff, L. M., and Schulz, E. metabench – a sparse benchmark of reasoning and knowledge in large language models. In *International Conference on Learning Representations (ICLR)*, 2025.

Li, Y., Ma, J., Ballesteros, M., Benajiba, Y., and Horwood, G. Active evaluation acquisition for efficient LLM benchmarking. In *International Conference on Machine Learning (ICML)*, 2025.

Liang, P., Bommasani, R., Lee, T., et al. Holistic evaluation of language models. *Transactions on Machine Learning Research*, 2023.

Lord, F. M. and Novick, M. R. *Statistical Theories of Mental Test Scores*. Addison-Wesley, 1968.

Maia Polo, F., Weber, L., Choshen, L., Sun, Y., Xu, G., and Yurochkin, M. tinybenchmarks: evaluating LLMs with fewer examples. In *International Conference on Machine Learning (ICML)*, 2024.

Pacchiardi, L., Cheke, L. G., and Hernández-Orallo, J. 100 instances is all you need: predicting the success of a new LLM on unseen data by testing on a few instances. *arXiv preprint arXiv:2409.03563*, 2024.

Reckase, M. D. *Multidimensional Item Response Theory*. Springer, 2009.

Särndal, C.-E., Swensson, B., and Wretman, J. *Model Assisted Survey Sampling*. Springer, 1992.

Vivek, R., Ethayarajh, K., Yang, D., and Kiela, D. Anchor points: Benchmarking models with much fewer examples. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, 2024.

Wang, Y., Ma, X., Zhang, G., Ni, Y., Chandra, A., Guo, S., Ren, W., Arulraj, A., He, X., Jiang, Z., Li, T., Ku, M., Wang, K., Zhuang, A., Fan, R., Yue, X., and Chen, W. MMLU-Pro: A more robust and challenging multi-task language understanding benchmark. In *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*, 2024.

Zhang, G., Dorner, F. E., and Hardt, M. How benchmark prediction from fewer data misses the mark. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

A. Proof of Proposition 1 and variance estimate

Fix a target t and drop the index. Let S be a uniformly random k-subset of $[n]$ and let $(S_1, \dots, S_K)$ be a uniformly random ordered partition of S into parts of fixed sizes $k_1, \dots, k_K$ (sizes depend on k and K only). Then (T_j, S_j), with $T_j = S \setminus S_j$, is uniformly distributed over ordered pairs of disjoint subsets of sizes $(k - k_j, k_j)$. Conditional on T_j, hence, S_j is a uniformly random k_j-subset of $C_j = [n] \setminus T_j$, $|C_j| = n - k + k_j$. The predictions $\hat{p}_i^{(-j)}$ are a deterministic function of T_j, the outcomes y_{T_j} and the source matrix X (nothing from S_j enters the ability estimate, and X is independent of the target's draw). Write $g_i = y_i - \hat{p}_i^{(-j)}$ for $i \in C_j$. Then

$$\mathbb{E}\Big[\tfrac{1}{k_j} \sum_{i \in S_j} g_i \,\Big|\, T_j\Big] = \frac{1}{|C_j|} \sum_{i \in C_j} g_i,$$

and therefore

$$\mathbb{E}[\hat{\theta}^{(j)} \mid T_j] = \frac{1}{n}\Big[\sum_{i \in T_j} y_i + \sum_{i \in C_j} \hat{p}_i^{(-j)} + |C_j| \cdot \frac{1}{|C_j|} \sum_{i \in C_j} (y_i - \hat{p}_i^{(-j)})\Big] = \frac{1}{n} \sum_{i=1}^{n} y_i = \theta.$$

Taking expectations over T_j and averaging over j gives $\mathbb{E}[\hat{\theta}^{\mathrm{CF}}] = \theta$. □

The proposition says nothing about variance: $\mathrm{Var}(\hat{\theta}^{\mathrm{CF}}) \approx (1 - k/n)\, s_r^2/k$ where s_r^2 is the item-level variance of the out-of-fold residuals $y_i - \hat{p}_i$, to be compared with s_y^2/k for the mean. For binary outcomes $s_r^2 \geq \overline{p(1-p)}$ (the Bernoulli noise given the true success probabilities), so gains saturate when the predictor is good. The Wald intervals in Section 4 use $\hat{s}_r$ and a finite-population correction.

B. Data

Source: `https://github.com/socialfoundations/benchmark-prediction`, commit `55faff4`, files `data/scores/*.jbl` (and `*_models.jbl` model lists for the Open LLM Leaderboard benchmarks), MIT licence; upstream sources are the Open LLM Leaderboard v2 and HELM. The pickles were converted to `.npz` with a restricted unpickler that only admits numpy/builtin types (`code/prep_data.py`). Duplicate model names (which have identical rows) were removed (first occurrence kept).

Table 3. Benchmarks after de-duplication. "dup.-free models" = models used; acc. = mean accuracy over models; range = min–max model accuracy; "const." = share of items answered identically (all correct or all wrong) by every model.

Benchmark	rows	models	items	acc.%	range %	const.
ARC-C	218	212	1172	52.3	20.5–75.1	2.7%
BBH	446	427	5761	44.7	24.1–74.7	0.0%
GPQA	447	428	1192	28.9	22.5–43.7	0.0%
IFEval	448	429	541	33.1	3.7–88.0	1.3%
MMLU-Pro	448	429	12032	28.1	10.3–59.7	0.0%
MuSR	448	429	756	39.9	32.3–53.6	1.3%
MATH-L5	447	428	894	5.0	0.0–35.6	13.3%
H:Commonsense	83	83	500	82.3	22.2–97.2	0.0%
H:GSM8K	83	83	1000	65.7	2.8–95.6	0.3%
H:LegalBench	79	79	2047	53.3	22.8–70.2	2.6%
H:MATH	83	83	437	56.8	2.5–93.4	0.2%
H:MedQA	83	83	1000	59.4	21.5–86.9	0.2%
H:MMLU	83	83	567	61.3	22.9–81.1	0.2%

C. Implementation details

IRT model. $z_{ti} = \beta_i + \alpha_i^\top \vartheta_t$ with MAP fit by L-BFGS-B (300 iterations; analytic gradients) on the source matrix; priors $\vartheta_s \sim \mathcal{N}(0, I)$ and $\alpha_i, \beta_i \sim \mathcal{N}(0, 1/0.5)$; initialisation from the SVD of the centred source matrix. The target ability is a Newton MAP estimate with a $\mathcal{N}(0, I)$ prior (8 steps, step clipping at 3). $d \in \{1, 2, 4, 8\}$ is selected per split by the held-out log-loss on unobserved items of a random 25% of the source models given 50 random observed items. gp-IRT's $\hat{b}^2$ follows

the recipe of Maia Polo et al. (2024) (fit on 75% of the sources, held-out source models observe a random half of the items, predict the other half, squared mean absolute error of the predicted accuracy) and $\hat{\sigma}^2$ is the mean of $\bar{y}_s(1-\bar{y}_s)$ over source models. Because $\hat{b}^2$ is estimated by an IID hold-out of source models, it is small (median $\hat{b}^2 \approx 5 \times 10^{-4}$ across splits, similar in all regimes), giving a median $\lambda \approx 0.11$–0.15 at $k = 50$: gp-IRT puts about 85–90% of its weight on p-IRT, and under extrapolation this weight is not adapted to the actual bias. The selected latent dimension was $d = 1, 2, 4, 8$ in 31%, 23%, 30%, 16% of splits (larger d on BBH and MMLU-Pro). **Anchors.** k-means (one initialisation, 50 iterations, scikit-learn) on a 50-dim SVD embedding of the (centred) item response vectors (exact SVD when $n < 2000$, otherwise randomised); representative = item closest to the centroid; weight = cluster share. Anchor-IRT clusters (α_i, β_i). **D-optimal.** Greedy maximisation of $\log\det(I + \sum_{i\in S} w_i \alpha_i \alpha_i^\top)$ with w_i the average of $p(1-p)$ over 60 source abilities (rank-one updates). **Ridge / kNN.** `RidgeCV` (11 penalties from 10^{-2} to 10^3) from the k outcomes to the full-benchmark accuracy of source models; kNN uses the 10 nearest source models in Hamming distance. **Recalibrated predictor.** Per-target logistic regression with offset β_i and features $(1, \beta_i, \alpha_i)$, ridge penalties $(0.1, 1, 1)$, 10 Newton steps; used in p-IRT (same combination rule), CF and CF-EB0. **CF.** $K = 5$ folds; for the empirical-Bayes step v_t is the fold-corrected variance of the out-of-fold residuals divided by k; the EB1 trend uses precision-weighted least squares of $\hat{\delta}_t$ on $\hat{\theta}_t^{\text{plug}}$ when $T \geq 6$.

D. Full results

Tables 4 to 7 give the relative RMSE for all k and all 29 (selector, estimator) combinations, Tables 8 to 11 give all metrics at $k = 50$, and Table 12 the Spearman correlation for all k.

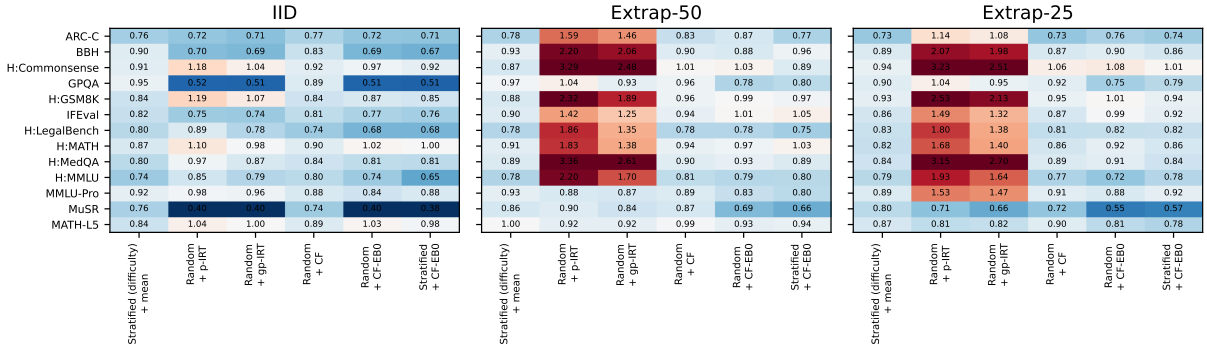


Figure 4. Per-benchmark RMSE ratios to Random+mean ($k = 50$; mean over 30 splits). Blue: better than random.

E. Ablation tables

All ablation numbers are relative RMSE at $k = 50$ against Random+mean computed on the *same* random subsets (geometric mean over 13 benchmarks of replicate-averaged RMSE; 10 replicate splits, so they differ slightly from the 30-split main results which use different draws).

Table 4. Relative RMSE vs. subset size k, IID (geometric mean over benchmarks; Random+mean = 1).

Method	k=10	25	50	100	200
Random + mean	1	1	1	1	1
Stratified (difficulty) + mean	0.88	0.85	0.84	0.84	0.83
Anchor points + weighted mean	0.89	0.98	1.13	1.32	1.51
Anchor-IRT + weighted mean	0.90	0.89	0.90	0.95	0.99
D-optimal + mean	1.61	2.03	2.51	3.23	4.02
Max-variance + mean	1.79	2.34	2.95	3.86	4.95
Random + ridge regression	0.56	0.64	0.71	0.78	0.86
Random + kNN over models	0.63	0.82	1.01	1.35	1.92
Random + p-IRT	0.72	0.80	0.83	0.88	0.92
Random + gp-IRT	0.71	0.77	0.78	0.80	0.82
Anchor-IRT + gp-IRT (tinyBench.-style)	0.65	0.68	0.71	0.77	0.86
D-optimal + p-IRT	0.70	0.85	1.00	1.16	1.21
Random + CF	0.90	0.86	0.83	0.82	0.80
Stratified + CF	0.89	0.84	0.82	0.82	0.79
Random + CF-EB0	0.66	0.72	0.75	0.79	0.81
Stratified + CF-EB0	0.65	0.71	0.73	0.78	0.81
Random + CF-EB1	0.81	0.75	0.74	0.76	0.79
Anchor points + p-IRT	0.66	0.73	0.78	0.96	1.44
Stratified + CF-EB1	0.78	0.72	0.72	0.76	0.78
Random + cross-fit plug-in	0.76	0.84	0.88	0.93	0.98
Random + p-IRT (recalibrated)	0.85	0.85	0.86	0.84	0.82
Stratified + p-IRT (recal.)	0.82	0.83	0.83	0.82	0.81
Anchor points + p-IRT (recal.)	0.95	1.07	1.11	1.04	1.32
Anchor-IRT + p-IRT (recal.)	0.77	0.82	0.87	0.91	1.04
D-optimal + p-IRT (recal.)	1.25	1.52	1.71	1.72	1.84
Max-variance + p-IRT (recal.)	1.32	1.68	1.91	2.07	1.81
Random + CF (recal.)	0.92	0.87	0.84	0.82	0.80
Stratified + CF (recal.)	0.89	0.84	0.82	0.82	0.79
Random + CF-EB0 (recal.)	0.84	0.84	0.83	0.82	0.81
Stratified + CF-EB0 (recal.)	0.81	0.82	0.82	0.82	0.79

Table 5. Relative RMSE vs. subset size k, Extrap-50 (geometric mean over benchmarks; Random+mean = 1).

Method	k=10	25	50	100	200
Random + mean	1	1	1	1	1
Stratified (difficulty) + mean	0.87	0.87	0.88	0.86	0.88
Anchor points + weighted mean	1.22	1.43	1.69	1.88	2.23
Anchor-IRT + weighted mean	0.96	0.99	1.06	1.09	1.25
D-optimal + mean	1.51	2.01	2.70	3.41	4.67
Max-variance + mean	1.59	2.15	2.96	3.85	5.25
Random + ridge regression	1.08	1.42	1.82	2.25	3.24
Random + kNN over models	1.18	1.72	2.43	3.30	5.03
Random + p-IRT	1.08	1.37	1.66	1.94	2.34
Random + gp-IRT	1.04	1.26	1.42	1.50	1.53
Anchor-IRT + gp-IRT (tinyBench.-style)	0.96	1.10	1.22	1.25	1.40
D-optimal + p-IRT	1.12	1.53	2.02	2.52	3.36
Random + CF	0.93	0.88	0.90	0.86	0.88
Stratified + CF	0.88	0.88	0.88	0.84	0.85
Random + CF-EB0	0.80	0.83	0.88	0.87	0.91
Stratified + CF-EB0	0.74	0.81	0.86	0.85	0.87
Random + CF-EB1	0.84	0.79	0.86	0.88	0.94
Anchor points + p-IRT	1.13	1.48	1.93	2.32	2.96
Stratified + CF-EB1	0.76	0.79	0.84	0.86	0.91
Random + cross-fit plug-in	1.11	1.42	1.73	2.04	2.53
Random + p-IRT (recalibrated)	0.87	0.91	0.96	0.95	0.99
Stratified + p-IRT (recal.)	0.83	0.88	0.94	0.95	0.98
Anchor points + p-IRT (recal.)	1.02	1.08	1.13	1.12	1.43
Anchor-IRT + p-IRT (recal.)	0.82	0.86	0.95	0.98	1.17
D-optimal + p-IRT (recal.)	1.19	1.53	1.91	2.05	2.20
Max-variance + p-IRT (recal.)	1.17	1.55	2.03	2.50	2.65
Random + CF (recal.)	0.92	0.89	0.89	0.84	0.85
Stratified + CF (recal.)	0.88	0.87	0.87	0.84	0.85
Random + CF-EB0 (recal.)	0.85	0.86	0.87	0.84	0.86
Stratified + CF-EB0 (recal.)	0.81	0.84	0.87	0.84	0.85

Table 6. Relative RMSE vs. subset size k, Extrap-25 (geometric mean over benchmarks; Random+mean = 1).

Method	k=10	25	50	100	200
Random + mean	1	1	1	1	1
Stratified (difficulty) + mean	0.87	0.85	0.85	0.85	0.83
Anchor points + weighted mean	1.02	1.25	1.45	1.75	1.96
Anchor-IRT + weighted mean	0.89	0.94	0.96	1.02	1.06
D-optimal + mean	1.53	2.14	2.79	3.75	4.98
Max-variance + mean	1.52	1.98	2.52	3.29	4.30
Random + ridge regression	0.95	1.22	1.41	1.72	2.12
Random + kNN over models	1.03	1.47	1.92	2.66	3.82
Random + p-IRT	1.15	1.44	1.61	1.83	2.00
Random + gp-IRT	1.12	1.34	1.42	1.45	1.39
Anchor-IRT + gp-IRT (tinyBench.-style)	1.04	1.14	1.15	1.17	1.16
D-optimal + p-IRT	1.08	1.43	1.74	2.13	2.67
Random + CF	0.89	0.87	0.86	0.84	0.82
Stratified + CF	0.87	0.84	0.84	0.83	0.81
Random + CF-EB0	0.78	0.83	0.84	0.85	0.85
Stratified + CF-EB0	0.76	0.80	0.83	0.84	0.82
Random + CF-EB1	0.78	0.77	0.79	0.84	0.88
Anchor points + p-IRT	1.15	1.45	1.67	1.91	2.55
Stratified + CF-EB1	0.76	0.72	0.77	0.82	0.86
Random + cross-fit plug-in	1.20	1.52	1.71	1.96	2.20
Random + p-IRT (recalibrated)	0.87	0.93	0.96	0.96	0.97
Stratified + p-IRT (recal.)	0.84	0.90	0.95	0.99	0.96
Anchor points + p-IRT (recal.)	1.10	1.23	1.17	1.07	1.23
Anchor-IRT + p-IRT (recal.)	0.82	0.94	1.01	1.04	1.13
D-optimal + p-IRT (recal.)	1.16	1.59	1.91	2.21	2.40
Max-variance + p-IRT (recal.)	1.16	1.49	1.83	2.11	1.95
Random + CF (recal.)	0.90	0.88	0.86	0.83	0.82
Stratified + CF (recal.)	0.88	0.85	0.85	0.84	0.81
Random + CF-EB0 (recal.)	0.84	0.86	0.84	0.83	0.82
Stratified + CF-EB0 (recal.)	0.81	0.83	0.83	0.83	0.80

Table 7. Relative RMSE vs. subset size k, Family hold-out (7 benchmarks) (geometric mean over benchmarks; Random+mean = 1).

Method	k=10	25	50	100	200
Random + mean	1	1	1	1	1
Stratified (difficulty) + mean	0.89	0.85	0.86	0.85	0.85
Anchor points + weighted mean	0.94	1.02	1.18	1.39	1.65
Anchor-IRT + weighted mean	0.91	0.89	0.90	0.90	0.96
D-optimal + mean	1.70	2.24	2.83	3.68	4.62
Max-variance + mean	1.95	2.54	3.25	4.22	5.73
Random + ridge regression	0.46	0.53	0.60	0.66	0.73
Random + kNN over models	0.48	0.58	0.70	0.88	1.23
Random + p-IRT	0.57	0.65	0.69	0.73	0.80
Random + gp-IRT	0.57	0.64	0.68	0.71	0.76
Anchor-IRT + gp-IRT (tinyBench.-style)	0.54	0.61	0.65	0.69	0.79
D-optimal + p-IRT	0.58	0.71	0.83	0.96	0.86
Random + CF	0.91	0.85	0.82	0.81	0.81
Stratified + CF	0.90	0.85	0.83	0.81	0.81
Random + CF-EB0	0.58	0.64	0.69	0.73	0.78
Stratified + CF-EB0	0.57	0.64	0.69	0.73	0.77
Random + CF-EB1	0.77	0.69	0.69	0.70	0.77
Anchor points + p-IRT	0.50	0.56	0.62	0.79	1.27
Stratified + CF-EB1	0.76	0.69	0.69	0.70	0.75
Random + cross-fit plug-in	0.59	0.67	0.72	0.76	0.83
Random + p-IRT (recalibrated)	0.87	0.87	0.87	0.90	0.87
Stratified + p-IRT (recal.)	0.84	0.84	0.84	0.85	0.87
Anchor points + p-IRT (recal.)	1.06	1.32	1.33	1.15	1.36
Anchor-IRT + p-IRT (recal.)	0.77	0.89	0.94	0.97	1.04
D-optimal + p-IRT (recal.)	1.30	1.70	1.80	1.76	1.99
Max-variance + p-IRT (recal.)	1.42	1.84	2.12	2.36	2.08
Random + CF (recal.)	0.91	0.88	0.83	0.82	0.81
Stratified + CF (recal.)	0.89	0.85	0.84	0.81	0.81
Random + CF-EB0 (recal.)	0.85	0.85	0.83	0.82	0.83
Stratified + CF-EB0 (recal.)	0.82	0.82	0.83	0.82	0.82

Table 8. All metrics at $k = 50$, IID: relative RMSE [95% CI], MAE (points), Spearman, Kendall, mean signed error (points); averages over benchmarks and 30 splits.

Method	rel. RMSE	MAE	Spearman	Kendall	bias
Random + mean	1 (ref.)	4.65	0.858	0.722	-0.26
Stratified (difficulty) + mean	0.84 [0.80, 0.88]	3.81	0.864	0.729	-0.00
Anchor points + weighted mean	1.13 [0.96, 1.32]	5.63	0.859	0.735	+0.25
Anchor-IRT + weighted mean	0.90 [0.85, 0.95]	4.09	0.864	0.717	-0.06
D-optimal + mean	2.51 [1.93, 3.34]	12.42	0.893	0.759	+6.27
Max-variance + mean	2.95 [2.17, 4.24]	15.00	0.917	0.778	+1.81
Random + ridge regression	0.71 [0.58, 0.84]	3.21	0.904	0.757	+0.06
Random + kNN over models	1.01 [0.73, 1.37]	4.72	0.884	0.730	+1.79
Random + p-IRT	0.83 [0.69, 0.97]	3.97	0.892	0.751	-0.73
Random + gp-IRT	0.78 [0.67, 0.90]	3.69	0.895	0.754	-0.62
Anchor-IRT + gp-IRT (tinyBench.-style)	0.71 [0.62, 0.80]	3.24	0.904	0.764	-0.44
D-optimal + p-IRT	1.00 [0.74, 1.34]	4.42	0.915	0.776	-0.09
Random + CF	0.83 [0.80, 0.87]	3.78	0.867	0.723	-0.00
Stratified + CF	0.82 [0.78, 0.86]	3.71	0.871	0.727	+0.01
Random + CF-EB0	0.75 [0.64, 0.85]	3.45	0.892	0.751	-0.33
Stratified + CF-EB0	0.73 [0.63, 0.84]	3.37	0.897	0.757	-0.29
Random + CF-EB1	0.74 [0.66, 0.81]	3.29	0.892	0.752	-0.12
Anchor points + p-IRT	0.78 [0.65, 0.93]	3.69	0.910	0.777	-0.54
Stratified + CF-EB1	0.72 [0.65, 0.80]	3.24	0.898	0.757	-0.10
Random + cross-fit plug-in	0.88 [0.72, 1.05]	4.27	0.891	0.750	-0.85
Random + p-IRT (recalibrated)	0.86 [0.81, 0.93]	3.79	0.862	0.718	+0.01
Stratified + p-IRT (recal.)	0.83 [0.78, 0.88]	3.69	0.870	0.725	-0.02
Anchor points + p-IRT (recal.)	1.11 [0.86, 1.63]	5.21	0.839	0.700	+1.46
Anchor-IRT + p-IRT (recal.)	0.87 [0.77, 1.00]	3.75	0.868	0.725	+0.13
D-optimal + p-IRT (recal.)	1.71 [1.35, 2.41]	7.81	0.829	0.674	+3.52
Max-variance + p-IRT (recal.)	1.91 [1.44, 2.70]	8.94	0.863	0.699	+1.16
Random + CF (recal.)	0.84 [0.81, 0.88]	3.81	0.862	0.718	-0.04
Stratified + CF (recal.)	0.82 [0.78, 0.86]	3.73	0.869	0.724	-0.02
Random + CF-EB0 (recal.)	0.83 [0.80, 0.87]	3.75	0.863	0.719	-0.03
Stratified + CF-EB0 (recal.)	0.82 [0.77, 0.86]	3.67	0.870	0.726	-0.01

Table 9. All metrics at $k = 50$, Extrap-50: relative RMSE [95% CI], MAE (points), Spearman, Kendall, mean signed error (points); averages over benchmarks and 30 splits.

Method	rel. RMSE	MAE	Spearman	Kendall	bias
Random + mean	1 (ref.)	4.58	0.733	0.575	+0.16
Stratified (difficulty) + mean	0.88 [0.84, 0.93]	3.96	0.737	0.577	-0.04
Anchor points + weighted mean	1.69 [1.29, 2.21]	8.97	0.706	0.534	-5.76
Anchor-IRT + weighted mean	1.06 [0.92, 1.30]	4.66	0.701	0.534	-0.33
D-optimal + mean	2.70 [2.08, 3.45]	15.28	0.588	0.451	+14.50
Max-variance + mean	2.96 [2.28, 4.02]	17.41	0.599	0.453	+16.28
Random + ridge regression	1.82 [1.50, 2.22]	9.30	0.518	0.379	-9.22
Random + kNN over models	2.43 [1.86, 3.15]	13.36	0.377	0.267	-13.36
Random + p-IRT	1.66 [1.29, 2.13]	9.12	0.545	0.414	-8.60
Random + gp-IRT	1.42 [1.15, 1.75]	7.49	0.666	0.508	-6.92
Anchor-IRT + gp-IRT (tinyBench.-style)	1.22 [1.05, 1.42]	5.87	0.699	0.532	-5.01
D-optimal + p-IRT	2.02 [1.60, 2.51]	10.49	0.524	0.395	-8.69
Random + CF	0.90 [0.86, 0.94]	4.04	0.734	0.565	+0.05
Stratified + CF	0.88 [0.84, 0.93]	3.96	0.735	0.565	-0.01
Random + CF-EB0	0.88 [0.82, 0.94]	3.98	0.727	0.559	-1.69
Stratified + CF-EB0	0.86 [0.80, 0.93]	3.92	0.728	0.560	-1.77
Random + CF-EB1	0.86 [0.77, 0.95]	3.87	0.655	0.498	-0.18
Anchor points + p-IRT	1.93 [1.52, 2.39]	10.20	0.497	0.371	-9.02
Stratified + CF-EB1	0.84 [0.74, 0.93]	3.77	0.649	0.494	-0.27
Random + cross-fit plug-in	1.73 [1.32, 2.26]	9.66	0.540	0.410	-9.12
Random + p-IRT (recalibrated)	0.96 [0.86, 1.12]	4.19	0.701	0.540	-0.51
Stratified + p-IRT (recal.)	0.94 [0.84, 1.10]	4.14	0.709	0.546	-0.68
Anchor points + p-IRT (recal.)	1.13 [0.93, 1.39]	5.28	0.690	0.529	+0.39
Anchor-IRT + p-IRT (recal.)	0.95 [0.83, 1.11]	4.20	0.722	0.553	-0.42
D-optimal + p-IRT (recal.)	1.91 [1.55, 2.39]	9.65	0.535	0.394	+5.76
Max-variance + p-IRT (recal.)	2.03 [1.60, 2.69]	10.75	0.579	0.426	+7.97
Random + CF (recal.)	0.89 [0.85, 0.92]	3.96	0.735	0.565	+0.10
Stratified + CF (recal.)	0.87 [0.83, 0.92]	3.91	0.735	0.566	-0.10
Random + CF-EB0 (recal.)	0.87 [0.83, 0.91]	3.89	0.730	0.560	-0.26
Stratified + CF-EB0 (recal.)	0.87 [0.82, 0.92]	3.87	0.732	0.563	-0.46

Table 10. All metrics at $k = 50$, Extrap-25: relative RMSE [95% CI], MAE (points), Spearman, Kendall, mean signed error (points); averages over benchmarks and 30 splits.

Method	rel. RMSE	MAE	Spearman	Kendall	bias
Random + mean	1 (ref.)	4.60	0.655	0.507	+0.06
Stratified (difficulty) + mean	0.85 [0.81, 0.89]	3.80	0.658	0.507	+0.04
Anchor points + weighted mean	1.45 [1.21, 1.72]	7.61	0.599	0.445	-4.58
Anchor-IRT + weighted mean	0.96 [0.86, 1.10]	4.23	0.632	0.474	+0.14
D-optimal + mean	2.79 [2.14, 3.75]	17.06	0.480	0.362	+16.58
Max-variance + mean	2.52 [1.77, 3.66]	16.57	0.559	0.425	+15.02
Random + ridge regression	1.41 [1.18, 1.65]	7.16	0.514	0.378	-7.08
Random + kNN over models	1.92 [1.51, 2.41]	10.41	0.356	0.254	-10.41
Random + p-IRT	1.61 [1.26, 2.07]	8.60	0.513	0.384	-8.18
Random + gp-IRT	1.42 [1.14, 1.79]	7.47	0.590	0.442	-7.02
Anchor-IRT + gp-IRT (tinyBench.-style)	1.15 [0.97, 1.38]	5.83	0.622	0.465	-5.16
D-optimal + p-IRT	1.74 [1.41, 2.11]	8.99	0.430	0.319	-8.27
Random + CF	0.86 [0.81, 0.92]	3.81	0.653	0.495	+0.07
Stratified + CF	0.84 [0.80, 0.89]	3.75	0.653	0.493	+0.13
Random + CF-EB0	0.84 [0.77, 0.93]	3.80	0.652	0.493	-1.75
Stratified + CF-EB0	0.83 [0.76, 0.90]	3.74	0.650	0.491	-1.68
Random + CF-EB1	0.79 [0.69, 0.90]	3.55	0.574	0.430	-0.17
Anchor points + p-IRT	1.67 [1.33, 2.05]	8.81	0.496	0.365	-8.54
Stratified + CF-EB1	0.77 [0.68, 0.87]	3.46	0.569	0.425	-0.13
Random + cross-fit plug-in	1.71 [1.32, 2.23]	9.25	0.506	0.379	-8.84
Random + p-IRT (recalibrated)	0.96 [0.83, 1.21]	4.55	0.568	0.433	-1.13
Stratified + p-IRT (recal.)	0.95 [0.81, 1.20]	4.46	0.575	0.437	-1.11
Anchor points + p-IRT (recal.)	1.17 [0.93, 1.48]	5.35	0.563	0.423	+0.52
Anchor-IRT + p-IRT (recal.)	1.01 [0.80, 1.34]	4.75	0.570	0.433	-0.83
D-optimal + p-IRT (recal.)	1.91 [1.53, 2.43]	9.77	0.417	0.305	+6.59
Max-variance + p-IRT (recal.)	1.83 [1.40, 2.47]	9.69	0.484	0.359	+5.42
Random + CF (recal.)	0.86 [0.81, 0.90]	3.80	0.649	0.492	+0.15
Stratified + CF (recal.)	0.85 [0.80, 0.90]	3.74	0.651	0.490	-0.02
Random + CF-EB0 (recal.)	0.84 [0.79, 0.89]	3.74	0.647	0.489	-0.26
Stratified + CF-EB0 (recal.)	0.83 [0.78, 0.88]	3.65	0.647	0.488	-0.35

Table 11. All metrics at $k = 50$, Family hold-out (7 benchmarks): relative RMSE [95% CI], MAE (points), Spearman, Kendall, mean signed error (points); averages over benchmarks and 30 splits.

Method	rel. RMSE	MAE	Spearman	Kendall	bias
Random + mean	1 (ref.)	4.60	0.805	0.661	-0.14
Stratified (difficulty) + mean	0.86 [0.80, 0.91]	3.87	0.802	0.659	-0.02
Anchor points + weighted mean	1.18 [0.94, 1.52]	5.97	0.787	0.659	-1.56
Anchor-IRT + weighted mean	0.90 [0.85, 0.95]	4.04	0.804	0.646	-0.39
D-optimal + mean	2.83 [1.98, 4.75]	13.24	0.858	0.709	+10.91
Max-variance + mean	3.25 [2.09, 5.73]	15.73	0.905	0.758	+13.12
Random + ridge regression	0.60 [0.48, 0.76]	2.61	0.881	0.726	-0.03
Random + kNN over models	0.70 [0.52, 0.99]	2.96	0.859	0.701	-0.47
Random + p-IRT	0.69 [0.56, 0.87]	3.12	0.852	0.700	+0.31
Random + gp-IRT	0.68 [0.55, 0.85]	3.07	0.856	0.704	+0.28
Anchor-IRT + gp-IRT (tinyBench.-style)	0.65 [0.56, 0.76]	2.92	0.861	0.710	-0.21
D-optimal + p-IRT	0.83 [0.53, 1.63]	3.70	0.881	0.726	+0.73
Random + CF	0.82 [0.78, 0.88]	3.67	0.817	0.662	-0.04
Stratified + CF	0.83 [0.78, 0.88]	3.72	0.814	0.660	-0.06
Random + CF-EB0	0.69 [0.56, 0.87]	3.12	0.853	0.701	+0.36
Stratified + CF-EB0	0.69 [0.56, 0.85]	3.11	0.850	0.699	+0.35
Random + CF-EB1	0.69 [0.60, 0.80]	3.01	0.852	0.699	-0.24
Anchor points + p-IRT	0.62 [0.53, 0.73]	2.77	0.874	0.731	-0.16
Stratified + CF-EB1	0.69 [0.60, 0.79]	3.02	0.850	0.699	-0.26
Random + cross-fit plug-in	0.72 [0.57, 0.92]	3.29	0.851	0.699	+0.41
Random + p-IRT (recalibrated)	0.87 [0.78, 1.00]	3.72	0.815	0.659	+0.02
Stratified + p-IRT (recal.)	0.84 [0.78, 0.91]	3.69	0.815	0.661	-0.03
Anchor points + p-IRT (recal.)	1.33 [0.84, 2.90]	6.40	0.774	0.617	+2.60
Anchor-IRT + p-IRT (recal.)	0.94 [0.77, 1.21]	3.88	0.802	0.647	+0.05
D-optimal + p-IRT (recal.)	1.80 [1.15, 3.55]	7.82	0.754	0.585	+3.14
Max-variance + p-IRT (recal.)	2.12 [1.37, 4.05]	10.10	0.817	0.646	+6.72
Random + CF (recal.)	0.83 [0.79, 0.90]	3.70	0.816	0.661	-0.05
Stratified + CF (recal.)	0.84 [0.78, 0.90]	3.73	0.813	0.658	-0.06
Random + CF-EB0 (recal.)	0.83 [0.78, 0.90]	3.68	0.816	0.661	+0.01
Stratified + CF-EB0 (recal.)	0.83 [0.77, 0.89]	3.68	0.815	0.661	-0.00

Table 12. Spearman correlation vs. k (mean over benchmarks and splits), by regime.

IID Method	k=10	25	50	100	200
Random + mean	0.669	0.785	0.858	0.910	0.951
Stratified (difficulty) + mean	0.670	0.796	0.864	0.915	0.952
Anchor points + weighted mean	0.734	0.819	0.859	0.889	0.918
Anchor-IRT + weighted mean	0.690	0.798	0.864	0.911	0.947
D-optimal + mean	0.757	0.845	0.893	0.930	0.951
Max-variance + mean	0.778	0.871	0.917	0.947	0.964
Random + ridge regression	0.739	0.849	0.904	0.941	0.966
Random + kNN over models	0.714	0.824	0.884	0.915	0.933
Random + p-IRT	0.724	0.837	0.892	0.931	0.957
Random + gp-IRT	0.725	0.840	0.895	0.935	0.961
Anchor-IRT + gp-IRT (tinyBench.-style)	0.755	0.852	0.904	0.936	0.957
D-optimal + p-IRT	0.776	0.877	0.915	0.940	0.959
Random + CF	0.676	0.795	0.867	0.919	0.956
Stratified + CF	0.680	0.803	0.871	0.920	0.956
Random + CF-EB0	0.718	0.835	0.892	0.933	0.960
Stratified + CF-EB0	0.720	0.838	0.897	0.935	0.960
Random + CF-EB1	0.716	0.837	0.892	0.932	0.959
Anchor points + p-IRT	0.782	0.869	0.910	0.934	0.949
Stratified + CF-EB1	0.721	0.839	0.898	0.934	0.959
Random + cross-fit plug-in	0.723	0.837	0.891	0.930	0.955
Random + p-IRT (recalibrated)	0.668	0.792	0.862	0.918	0.956
Stratified + p-IRT (recal.)	0.676	0.800	0.870	0.921	0.956
Anchor points + p-IRT (recal.)	0.674	0.779	0.839	0.900	0.942
Anchor-IRT + p-IRT (recal.)	0.704	0.808	0.868	0.913	0.943
D-optimal + p-IRT (recal.)	0.699	0.772	0.829	0.884	0.923
Max-variance + p-IRT (recal.)	0.722	0.805	0.863	0.900	0.929
Random + CF (recal.)	0.669	0.790	0.862	0.918	0.956
Stratified + CF (recal.)	0.675	0.800	0.869	0.921	0.956
Random + CF-EB0 (recal.)	0.670	0.791	0.863	0.919	0.956
Stratified + CF-EB0 (recal.)	0.676	0.801	0.870	0.921	0.956
Extrap-50 Method	k=10	25	50	100	200
Random + mean	0.459	0.611	0.733	0.816	0.895
Stratified (difficulty) + mean	0.465	0.627	0.737	0.820	0.896
Anchor points + weighted mean	0.494	0.613	0.706	0.783	0.852
Anchor-IRT + weighted mean	0.449	0.601	0.701	0.785	0.870
D-optimal + mean	0.363	0.489	0.588	0.667	0.749
Max-variance + mean	0.365	0.497	0.599	0.674	0.744
Random + ridge regression	0.257	0.390	0.518	0.620	0.743
Random + kNN over models	0.170	0.282	0.377	0.437	0.511
Random + p-IRT	0.317	0.442	0.545	0.653	0.773
Random + gp-IRT	0.365	0.532	0.666	0.777	0.878
Anchor-IRT + gp-IRT (tinyBench.-style)	0.442	0.590	0.699	0.787	0.872
D-optimal + p-IRT	0.332	0.437	0.524	0.597	0.683
Random + CF	0.455	0.615	0.734	0.819	0.898
Stratified + CF	0.460	0.629	0.735	0.821	0.899
Random + CF-EB0	0.438	0.599	0.727	0.815	0.895
Stratified + CF-EB0	0.452	0.616	0.728	0.817	0.897
Random + CF-EB1	0.390	0.528	0.655	0.771	0.872
Anchor points + p-IRT	0.290	0.394	0.497	0.644	0.754
Stratified + CF-EB1	0.388	0.528	0.649	0.775	0.874
Random + cross-fit plug-in	0.317	0.441	0.540	0.640	0.757
Random + p-IRT (recalibrated)	0.450	0.580	0.701	0.796	0.884
Stratified + p-IRT (recal.)	0.447	0.605	0.709	0.798	0.884
Anchor points + p-IRT (recal.)	0.380	0.557	0.690	0.797	0.880
Anchor-IRT + p-IRT (recal.)	0.459	0.621	0.722	0.802	0.884
D-optimal + p-IRT (recal.)	0.329	0.448	0.535	0.628	0.744
Max-variance + p-IRT (recal.)	0.351	0.478	0.579	0.632	0.716
Random + CF (recal.)	0.455	0.611	0.735	0.819	0.898
Stratified + CF (recal.)	0.459	0.628	0.735	0.821	0.899
Random + CF-EB0 (recal.)	0.457	0.608	0.730	0.818	0.897
Stratified + CF-EB0 (recal.)	0.454	0.621	0.732	0.820	0.899
Extrap-25 Method	k=10	25	50	100	200
Random + mean	0.375	0.519	0.655	0.764	0.859
Stratified (difficulty) + mean	0.362	0.526	0.658	0.762	0.862
Anchor points + weighted mean	0.365	0.487	0.599	0.712	0.819
Anchor-IRT + weighted mean	0.355	0.513	0.632	0.744	0.834
D-optimal + mean	0.268	0.368	0.480	0.579	0.673
Max-variance + mean	0.302	0.429	0.559	0.652	0.737
Random + ridge regression	0.252	0.357	0.514	0.618	0.752
Random + kNN over models	0.164	0.247	0.356	0.414	0.495
Random + p-IRT	0.266	0.359	0.513	0.636	0.764
Random + gp-IRT	0.292	0.416	0.590	0.722	0.839
Anchor-IRT + gp-IRT (tinyBench.-style)	0.346	0.491	0.622	0.744	0.836
D-optimal + p-IRT	0.231	0.317	0.430	0.539	0.632
Random + CF	0.372	0.512	0.653	0.764	0.861
Stratified + CF	0.364	0.527	0.653	0.765	0.863
Random + CF-EB0	0.356	0.504	0.652	0.766	0.861
Stratified + CF-EB0	0.344	0.514	0.650	0.767	0.864
Random + CF-EB1	0.297	0.404	0.574	0.713	0.830
Anchor points + p-IRT	0.252	0.372	0.496	0.645	0.771
Stratified + CF-EB1	0.306	0.429	0.569	0.718	0.837

Table 13. Number of source models (IRT with $d = 2$; sources subsampled from the source pool).

	IID			Extrap-50		
m	p-IRT	CF	CF-EB0	p-IRT	CF	CF-EB0
8	1.43	0.89	0.84	1.93	0.95	0.90
16	1.22	0.87	0.82	1.82	0.92	0.89
32	1.03	0.85	0.78	1.79	0.92	0.90
64	0.82	0.84	0.71	1.33	0.92	0.88
128	0.72	0.83	0.67	1.33	0.91	0.85

Table 14. Latent dimension d and strength λ_ϑ of the ability prior used when estimating a target's ability (all other settings as in the main experiment).

	IID			Extrap-50		
setting	p-IRT	CF	CF-EB0	p-IRT	CF	CF-EB0
$d = 1$, $\lambda_\vartheta = 0.1$	0.72	0.82	0.69	1.62	0.89	0.91
$d = 1$, $\lambda_\vartheta = 1.0$	0.89	0.83	0.75	1.99	0.89	0.87
$d = 2$, $\lambda_\vartheta = 0.1$	0.68	0.82	0.67	1.47	0.90	0.91
$d = 2$, $\lambda_\vartheta = 1.0$	0.82	0.83	0.74	1.84	0.90	0.88
$d = 4$, $\lambda_\vartheta = 0.1$	0.69	0.83	0.69	1.31	0.92	0.96
$d = 4$, $\lambda_\vartheta = 1.0$	0.80	0.82	0.75	1.67	0.89	0.90
$d = 8$, $\lambda_\vartheta = 0.1$	0.74	0.88	0.74	1.21	0.99	0.98
$d = 8$, $\lambda_\vartheta = 1.0$	0.80	0.83	0.77	1.47	0.93	0.91

Table 15. Number of cross-fitting folds K (relative RMSE, $d = 2$).

	IID		Extrap-50	
	CF	CF-EB0	CF	CF-EB0
$K = 2$	0.84	0.74	0.90	0.88
$K = 5$	0.83	0.74	0.90	0.88
$K = 10$	0.82	0.74	0.90	0.87

Table 16. Recalibrated predictor and classical regression (GREG) estimators ($d = 4$): relative RMSE and Spearman ρ at $k = 50$. "Recalibrated": per-target logistic fit with free intercept and difficulty slope (Section 3).

	IID		Extrap-50	
Method	rel. RMSE	ρ	rel. RMSE	ρ
p-IRT	0.80	0.896	1.67	0.588
p-IRT, recalibrated	0.81	0.865	0.90	0.729
CF	0.82	0.863	0.89	0.731
CF, recalibrated predictor	0.83	0.863	0.90	0.731
CF-EB0	0.75	0.896	0.90	0.721
CF-EB0, recalibrated	0.81	0.864	0.88	0.732
Regression est. (source difficulty)	0.86	0.846	0.91	0.725
Regression est. (IRT item features)	0.86	0.861	0.93	0.727
Stratified + mean	0.83	0.857	0.91	0.750
Stratified + CF	0.82	0.863	0.90	0.751
Stratified + CF-EB0	0.74	0.897	0.87	0.743
Stratified + regression (difficulty)	0.83	0.856	0.90	0.748
Stratified + regression (IRT feat.)	0.84	0.861	0.90	0.745